

TOA- Assignment 2

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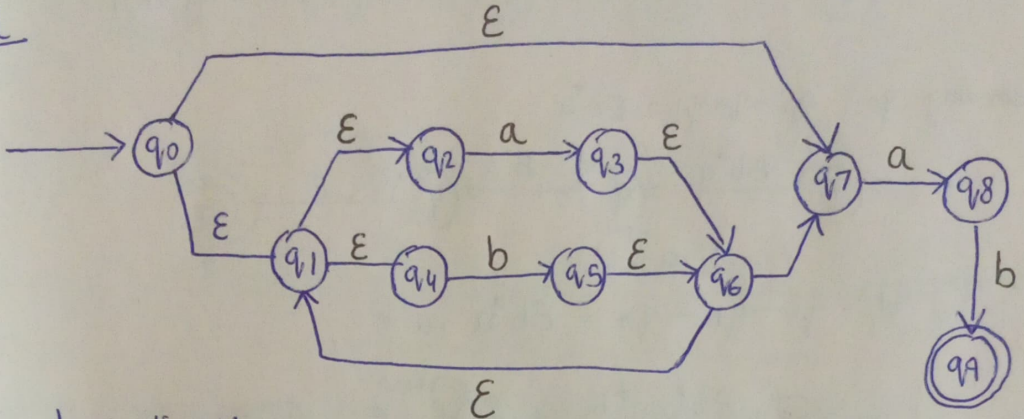
Class :- BCS-3E

pg 1

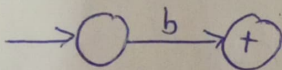
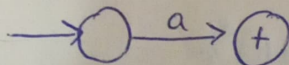
Section A :-

Q1a) Regular Expression :- $(a+b)^*ab$

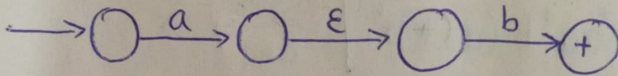
sol



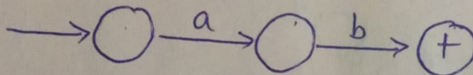
working of sub diagrams



so ab (concatenate)

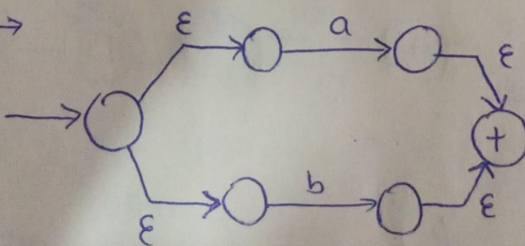


simplify



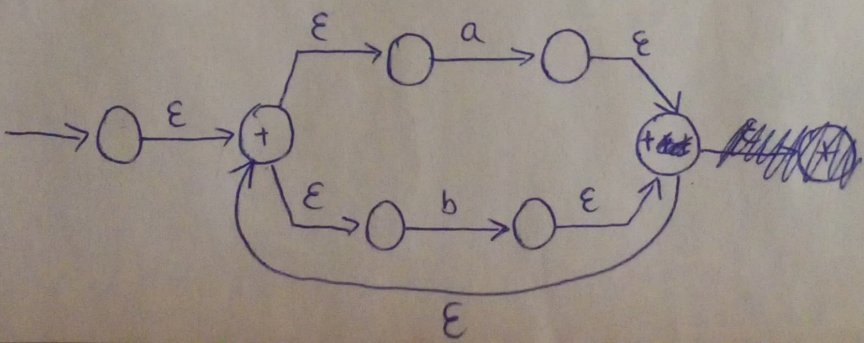
for $(a+b)^*$

$a+b \rightarrow$

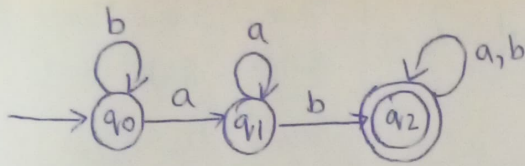


now

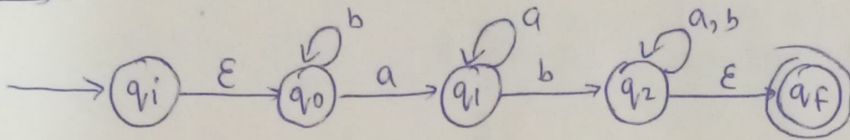
$(a+b)^*$



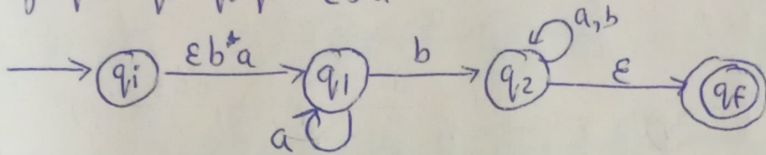
Q1b)



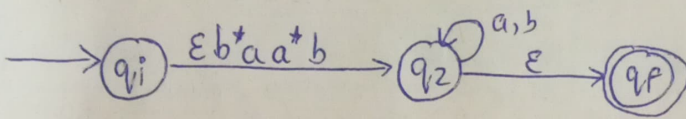
working



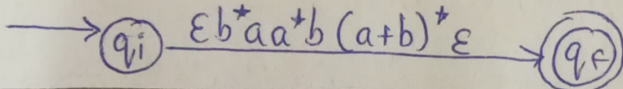
Removing q_0 : $q_i - q_0 q_1 = \epsilon b^* a$



Removing q_1 : $q_i - q_1 - q_2 = \epsilon b^* a \cdot a^* b$

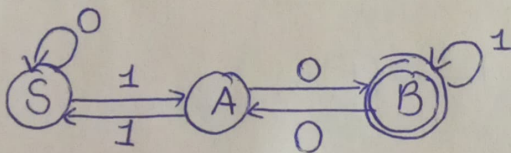


Removing q_2 : $q_i - q_2 - q_f = (\epsilon b^* a a^* b) \cdot (a+b)^* \epsilon$

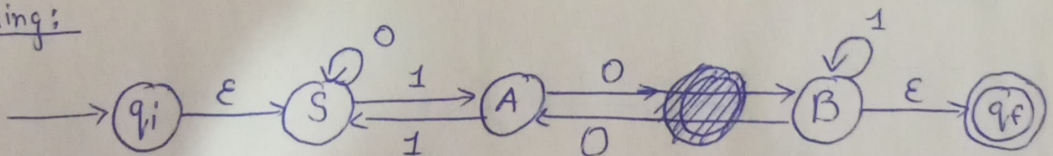


$\therefore$ Regular Expression : $b^* a a^* b (a+b)^*$ Ans

Q2a)



working:



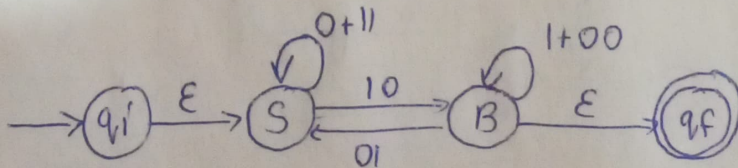
Removing A:

SAB : 10

BAS : 01

SAS : 11

BAB : 00

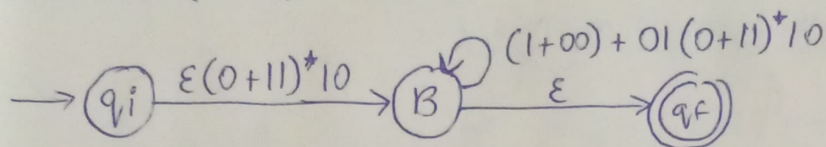


Removing S :

pg 3

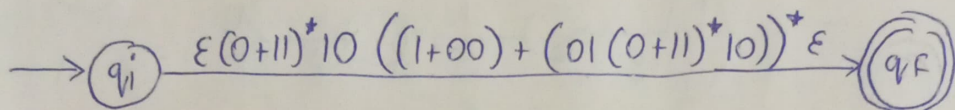
$$q_i - S - B = \epsilon(0+11)^*10$$

$$B - S - B = (1+00) + 01(0+11)^*10$$



Removing B :

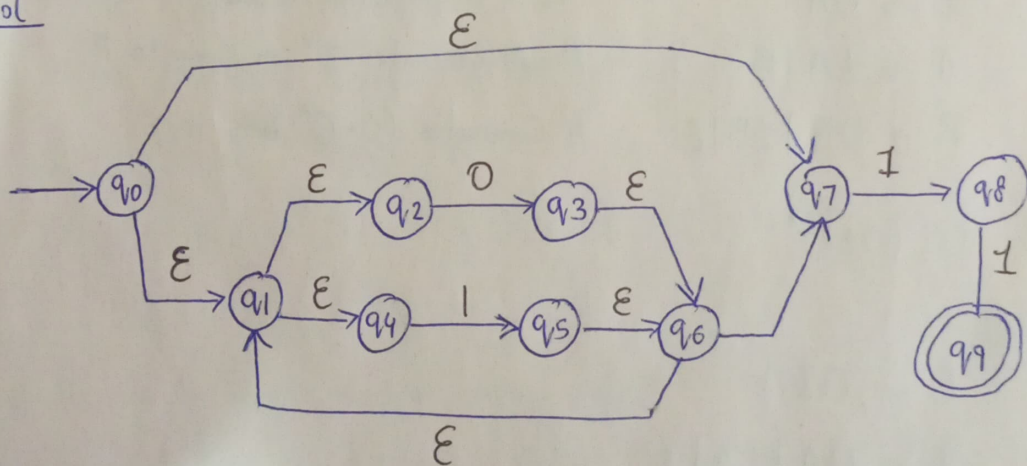
$$q_i - B - q_f = \epsilon(0+11)^*10((1+00) + (01(0+11)^*10))^*\epsilon$$



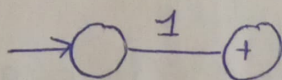
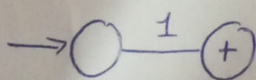
$\therefore$ Regular Expression : $(0+11)^*10((1+00) + (01(0+11)^*10))^*$ Ans

Q2b) Regular Expression :- $(0+1)^*11$

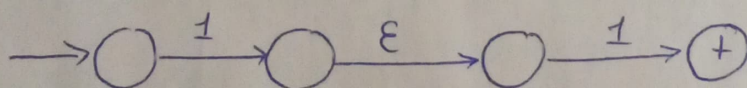
Sol



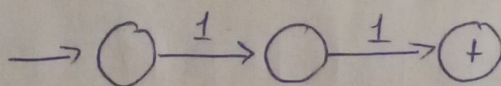
working of sub diagram



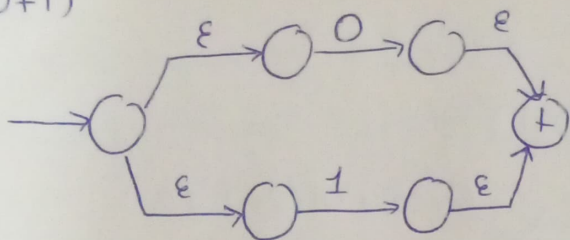
So 11 concatenation



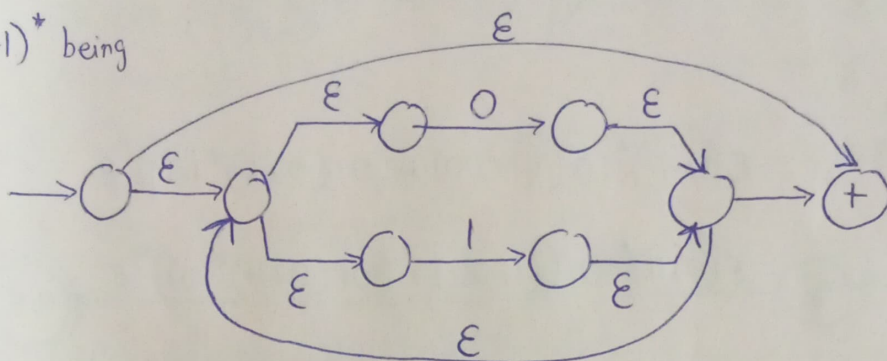
simplify



for $(0+1)$



$\therefore (0+1)^*$ being



Section B:

Q1i) $01^*[0+1]^*$

sol

$$S \rightarrow 0A$$

// will begin with 0 always

$$A \rightarrow 1A \mid B$$

// will generate 1^* or jump to B

$$B \rightarrow 0B \mid 1B \mid \epsilon$$

// Generates $(0+1)^*$ ends in ϵ

ii) $01(0+1)^*$

sol

$$S \rightarrow 01A$$

$$A \rightarrow 0A \mid 1A \mid \epsilon$$

iii) Letters followed by digits

$$L = \{xy \mid x \in \{a, \dots, z, A, \dots, Z\} \mid y \in \{0-9\}^*\}$$

my assumption string is of two length, and small + capital letters

sol

$$S \rightarrow aQ_0 \mid bQ_0 \mid \dots \mid zQ_0 \mid AQ_0 \mid BQ_0 \mid \dots \mid ZQ_0$$

$$Q_0 \rightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$$

iv) Start and end with a.

Regular expression: $a(a+b)^*a$

assuming Alphabet $\{a, b\}$

Sol

$$S \rightarrow aA|a$$

$$A \rightarrow aA|bA|a$$

// always start with a, will accept single a

// $(a+b)^*$ displays multiple time will always terminate at a

accepted string = (a, aaa, aababa)

v) Generate all even integer upto 998

eg strings (0, 2, 4, ..., 10, 12, 14, ..., 102, 104, ..., 312, 396, ..., 998)

Sol

$$S \rightarrow 0|X|NE|NDE$$

$$X \rightarrow 2|4|6|8$$

$$N \rightarrow 1|2|3|4|5|6|7|8|9$$

$$E \rightarrow 0|2|4|6|8$$

$$D \rightarrow 0|1|2|3|4|5|6|7|8|9$$

where

X is single digit number being printed

NE is double digit number (10, 12, 42, 46, 66, 78, 98)

NDE is triple digit number (106, 108, 612, 746, 908, 998)

vi) Alternate Letters

~~Assumpti~~ Assuming alphabet $\{a, b\}$

so Regular Expression: $(ab)^* + (ba)^*$

eg strings (ab, ababab, ba, baba, ...)

Sol

$$S \rightarrow A|B$$

$$A \rightarrow abA|\epsilon \quad // \text{ will give } (ab)^*$$

// will give $(ba)^*$

$$B \rightarrow baB|\epsilon$$

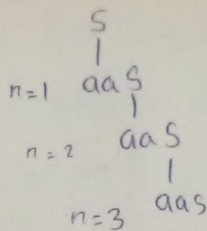
vii) a^{2n}

$$L = \{a^{2n} \mid n \geq 0\}$$

so Regular expression = $(aa)^*$

sol

$$S \rightarrow aaS \mid \epsilon$$



$S \rightarrow \epsilon \rightarrow n=0$

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viii) $[ab]^n$

$$L = \{(ab)^n \mid n \geq 0\}$$

so Regular expression = $(ab)^*$

sol

$$S \rightarrow abS \mid \epsilon$$

ix) Generate $\{012, 120, 201\}$

sol 1

$$S \rightarrow 012 \mid 120 \mid 201$$

sol 2

$$S \rightarrow 0A \mid 1B \mid 2C$$

$$A \rightarrow 1D$$

$$D \rightarrow 2$$

$$B \rightarrow 2E$$

$$E \rightarrow 0$$

$$C \rightarrow 0F$$

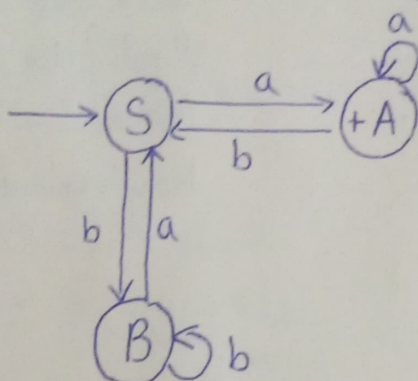
$$F \rightarrow 1$$

x) Grammar for the set of all strings over $\{0, 1\}$ consisting of equal number of 0's and 1's

$$S \rightarrow 0S1 \mid 1S0 \mid 10S \mid 01S \mid \epsilon$$

Q2a) $S \rightarrow aA \mid bB$
 $A \rightarrow aA \mid bS \mid \epsilon$
 $B \rightarrow bB \mid aS$

FA:



working:

Initial State : S

Non-terminal : {S, A, B}

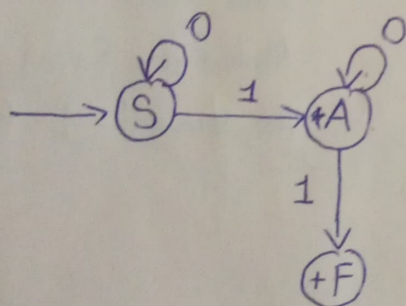
Terminal : {a, b}

Final : A as ϵ terminator

δ	a	b
S	A	B
A	A	S
B	S	B

b) $S \rightarrow 0S \mid 1A$
 $A \rightarrow 0A \mid 1$

Sol



working:

Initial State : S

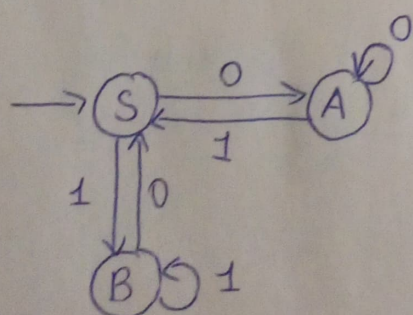
Final State : A as 1 terminate

Non-terminal : {A, S, F}

Alphabet/Terminal : {0, 1}

δ	0	1
S	S	A
A	A	F

c) $S \rightarrow 0A \mid 1B$
 $A \rightarrow 1S \mid 0A$
 $B \rightarrow 0S \mid 1B$



working:

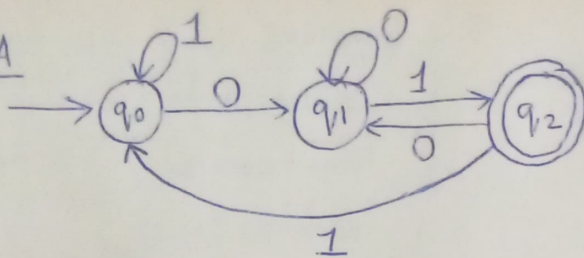
Initial State : S

Non-terminal State : {S, A, B}

Alphabet : {0, 1}

Final state :- none as no terminal or epsilon end, infinite loop

Q3a) DFA



Right-Linear Grammar [RLG]

$$Q_0 \rightarrow 0Q_1 \mid 1Q_0$$

$$Q_1 \rightarrow 0Q_1 \mid 1Q_2$$

$$Q_2 \rightarrow 0Q_1 \mid 1Q_0 \mid \epsilon$$

Start State : Q_0

Final State : Q_2

Alphabet = $\{0, 1\}$

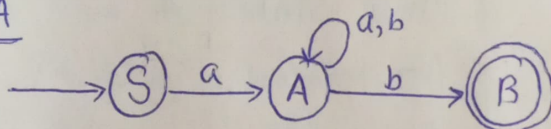
Non-terminal = $\{Q_0, Q_1, Q_2\}$

CFG

$$S \rightarrow X01$$

$$X \rightarrow 0X \mid 1X \mid \epsilon$$

b) NFA



Start State : S

Final State : B

Alphabet = $\{a, b\}$

Non-terminal = $\{S, A, B\}$

Right-Linear Grammar [RLG]

$$S \rightarrow aA$$

$$A \rightarrow aA \mid bA \mid bB$$

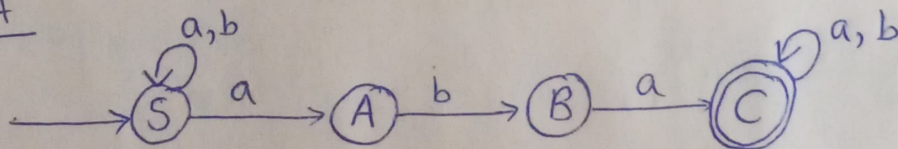
$$B \rightarrow \epsilon$$

CFG

$$S \rightarrow aMb$$

$$M \rightarrow aM \mid bM \mid \epsilon$$

c) NFA



Start State : S

Final State : C

Alphabet : $\{a, b\}$

Non-Terminal = $\{S, A, B, C\}$

S

a S } (S)^{a,b}

b S }

a A } (S) $\xrightarrow{a}$ (A)^{a,b}

b B } (S) $\xrightarrow{a}$ (A) $\xrightarrow{b}$ (B)^{a,b}

a C } (S) $\xrightarrow{a}$ (A) $\xrightarrow{b}$ (B) $\xrightarrow{c}$ (C)^{a,b}

a C } self loop on C
bc
E

aA }

$\{ \begin{matrix} | \\ b \end{matrix} B \}$

$a \in C \} \quad \overset{a,b}{\text{Path}} \quad S \xrightarrow{a} A \xrightarrow{b} B \xrightarrow{a} C$

ac ? self loop

bc } self loop on c

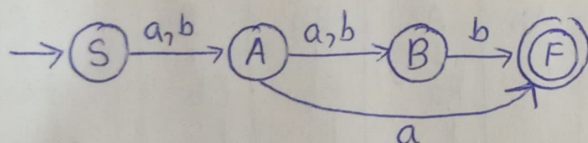
E (C)

Original LLC

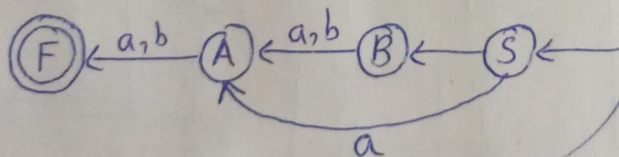
Find string produced by LLg

$$S \rightarrow Aa \mid Ab$$
$$A \rightarrow ba \mid bb \mid a$$
$$S \rightarrow baa \mid bba \mid aa \mid bab \mid bbb \mid ab$$
$$L = \{aa, baa, bba, ab, bab, bbb\}$$

generated



To check if RLG was correctly converted let's confirm using parser tree



$ \begin{array}{c} S \\ \\ bB \\ \\ aA \\ \\ aF \\ \\ \epsilon \end{array} $	$ \begin{array}{c} S \\ \\ bB \\ \\ bA \\ \\ aF \\ \\ \epsilon \end{array} $	$ \begin{array}{c} S \\ \\ bB \\ \\ aA \\ \\ bF \\ \\ \epsilon \end{array} $	$ \begin{array}{c} S \\ \\ bB \\ \\ bA \\ \\ bF \\ \\ \epsilon \end{array} $
baa	$= bba$	$= bab$	$= bbb$

$$\begin{array}{c} S \\ | \\ aA \\ | \\ qF \\ | \\ \varepsilon \end{array}$$
$$\begin{array}{c} S \\ | \\ aA \\ | \\ bF \\ | \\ E \end{array}$$
$$\begin{array}{c} bB \\ | \\ aA \\ | \\ a \\ 8 \end{array}$$
$$= bb$$
$$= b$$
$$= b$$
$$= aa$$

1-94

- ha

$$= bba$$

- bad

11

∴ The conversion was successful as strings made by LLG matches with strings made by RLG. You can see parse tree

Q4b) Convert the following RLG $\rightarrow$ LLG

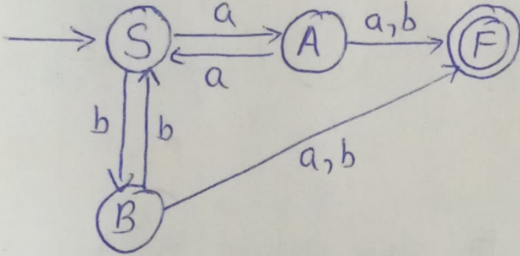
Original RLG

$S \rightarrow aA \mid bB$

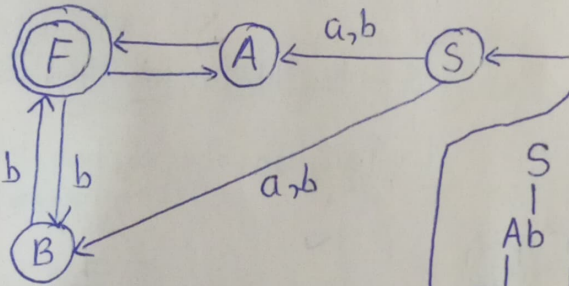
$A \rightarrow a \mid b \mid aS$

$B \rightarrow b \mid a \mid bS$

Create FA



Reverse FA



LLG:

$S \rightarrow Aa \mid Ab \mid Ba \mid Bb$

$A \rightarrow Fa$

$B \rightarrow Fb$

$F \rightarrow Aa \mid Bb \mid \epsilon$

if we simplify it then

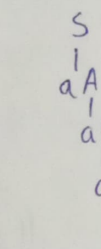
$S \rightarrow Aa \mid Ab \mid Ba \mid Bb$

$A \rightarrow a \mid Aaa \mid Bba$

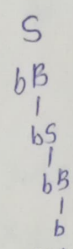
$B \rightarrow b \mid Aab \mid Bbb$

Find some strings made by RLG to confirm with LLG string

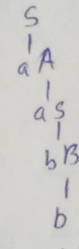
$\{aa, ab, ba, bb, aaaa, bbbb, aabb, aaba\}$



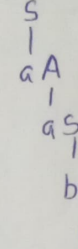
= aaaaa



= bbbbb



= aabb

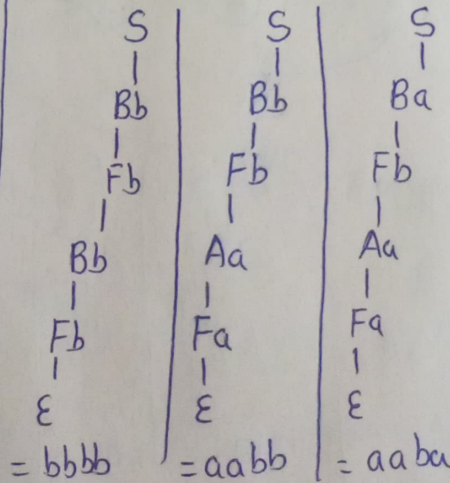
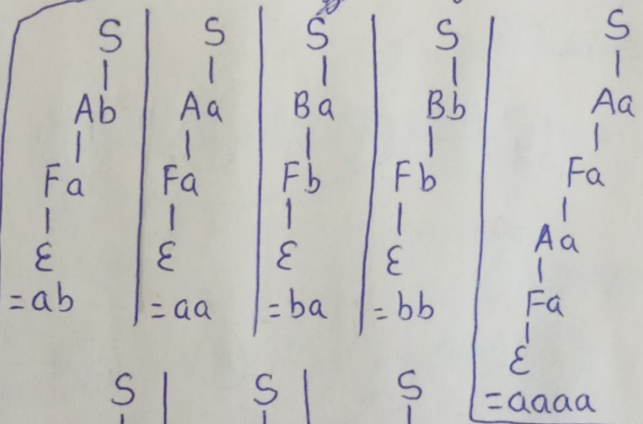


= aaba

$\therefore$ Test case string

aa, ab, ba, bb, aaaa, bbbb, aabb, aaba

To check if LLG was correctly converted, let's confirm using parse tree. (using simplified LLG)



$\therefore$ The conversion was successful as strings made by RLG matches with string made by LLG. You can see parse tree.

Q5) For 1

$$S \rightarrow aSb \mid ab$$

string = aabb

a) substitution

$$\begin{aligned} S &\rightarrow aSb \\ a(ab)b \\ aabb \end{aligned}$$

String achieved

b) Left most derivation

$$\begin{aligned} S &\rightarrow aSb \\ a(ab)b \\ aabb \end{aligned}$$

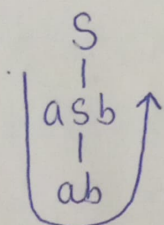
String achieved

Right most derivation

$$\begin{aligned} S &\rightarrow aSb \\ a(ab)b \\ aabb \end{aligned}$$

String achieved

c) Parse tree



= aabb

String achieved

Note = All three derivation coincide cause at each stage there is just one non-terminal

Q5) For 2

$$S \rightarrow aS \mid Sb \mid ab$$

string = aaabb

a) substitution

$$\begin{aligned} S &\rightarrow aS \\ a(aS) \\ aa(Sb) \\ aa(ab)b \\ aaabb \end{aligned}$$

String achieved

b) Left most derivation

$$\begin{aligned} S &\rightarrow Sb \\ (aS)b \\ a(aS)b \\ aa(ab)b \\ aaabb \end{aligned}$$

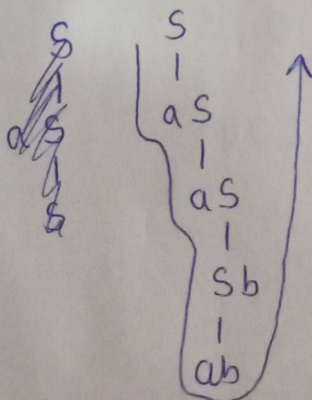
String achieved

Right most derivation

$$\begin{aligned} S &\rightarrow aS \\ a(aS) \\ aa(Sb) \\ aa(ab)b \\ aaabb \end{aligned}$$

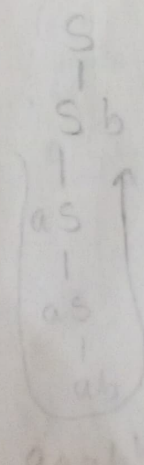
String achieved

c) Parse tree



= aaabb

String achieved



Q5) For 3

$S \rightarrow 0S1 | 01$ string 000111

a) Substitution

$S \rightarrow 0S1$

$0(0S1)1$

$00(01)11$

000111

String achieved

b) Left most derivation ; Right most derivation

$S \rightarrow 0S1$

$0(0S1)1$

$00(01)11$

000111

String achieved

$S \rightarrow 0S1$

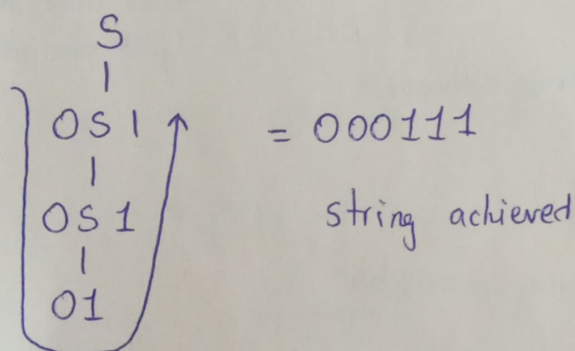
$0(0S1)1$

$00(01)11$

000111

String achieved

c) Parse Tree



Q6i) CGF for balanced parenthesis in this alphabet set $\{ (,) \}$

sol

$CGF \Rightarrow S \rightarrow (S) | SS | \epsilon$

eg string: $()$, $(())$, $(())(())$...

ii a) CGF of even palindrome 001100

sol

$CGF \Rightarrow S \rightarrow 0S0 | 1S1 | \epsilon$

eg string: 001100, 010010, 011110, ...

ii b) CGF of odd palindrome 00100

sol

$CGF \Rightarrow S \rightarrow 0S0 | 1S1 | 0 | 1$

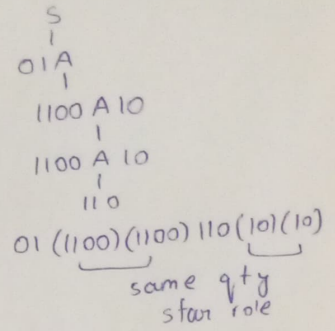
eg string: 01010, 01110, 00100, ...

iii) $\{01(1100)^n 110(10)^n \mid n \geq 0\}$

sol

$$S \rightarrow 01A$$

$$A \rightarrow 1100A10 \mid 110$$



iv) $\{0^n 1^m \mid n \neq m\}$

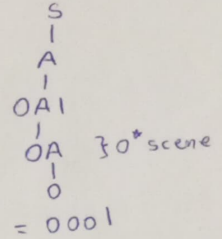
sol

so $n > m$ and $n < m$ means 0 more than 1 and 1 more than 0.

$$S \rightarrow A \mid B$$

$$A \rightarrow 0A1 \mid 0A \mid 0$$

$$B \rightarrow 0B1 \mid B1 \mid 1$$



eg string ; 0, 1, 001, 011, 0001, 0111, ...

v) $\{[0^n 1^n 2^m 3^m \mid n \geq 1, m \geq 1] \text{ or } [0^n 1^m 2^m 3^n \mid n \geq 1, m \geq 1]\}$

sol

Language 1

Language 2

union of it in $S \rightarrow L_1 \mid L_2$

for Language 1

$$A \rightarrow XY$$

$$X \rightarrow 0X1 \mid 01$$

$$Y \rightarrow 2Y3 \mid 23$$

// generates $0^n 1^n \quad n \geq 1$

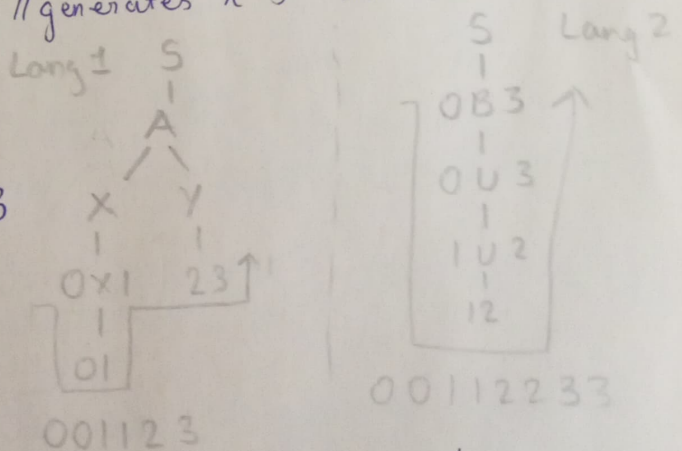
// generates $2^m 3^m \quad m \geq 1$

for Language 2

$$B \rightarrow \cancel{0B3} \mid 0U3$$

$$\cancel{01123} \mid \cancel{0123}$$

$$U \rightarrow 1U2 \mid 12$$



CFG

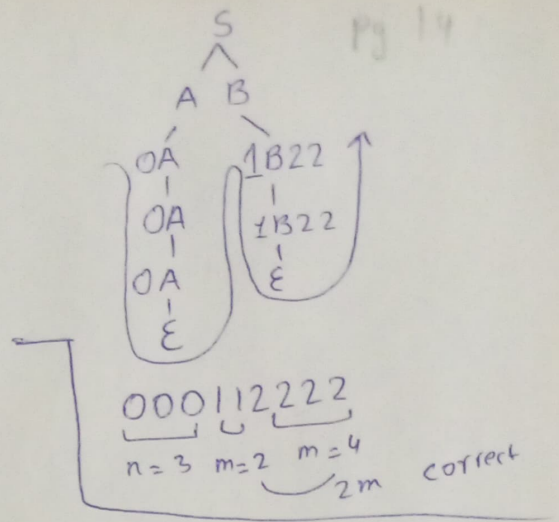
$$S \rightarrow A \mid B$$

// This will call language 1 grammar or language 2 grammar

Viii) $\{0^n 1^m 2^{2m} \mid n, m \geq 0\}$

Sol

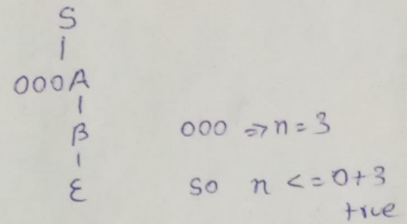
$$\begin{aligned} \text{CFG} \Rightarrow & S \rightarrow AB \\ & A \rightarrow 0A \mid \epsilon \\ & B \rightarrow 1B22 \mid \epsilon \end{aligned}$$



ix) $\{0^n 1^m \mid n \leq m+3\}$

$$S \rightarrow 000A \mid 00A \mid 0A \mid A$$

$$\begin{aligned} A &\rightarrow 0A1 \mid B \\ B &\rightarrow 1B \mid \epsilon \end{aligned} \quad \begin{aligned} A &\rightarrow 0A1 \mid B \\ B &\rightarrow 1B \mid \epsilon \end{aligned}$$



Q7e) Input Alphabet = $\{[,], \{, \}, (,)\}$

$$\text{CFG} \Rightarrow S \rightarrow (S) \mid [S] \mid \{S\} \mid SS \mid \epsilon$$

Section C

$$\begin{aligned} \text{CFG} \Rightarrow & S \rightarrow aSa \mid bSb \mid a \mid b \\ & A \rightarrow B \mid a \\ & B \rightarrow C \mid b \end{aligned}$$

a) Removing useless production from CFG

non-terminal states A, B, C are not reachable from S so removing them

$$\therefore S \rightarrow aSa \mid bSb \mid a \mid b$$

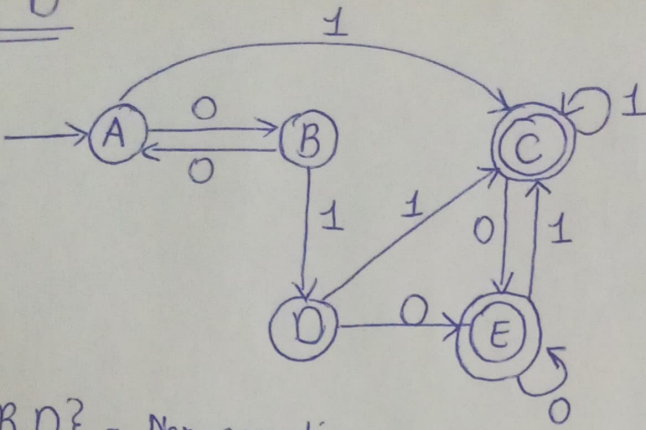
b) Check for ambiguity in the given CFG for the string 'abba'

The CFG generates odd length palindrome, while string "abba" is of even length. Therefore, it ain't in the language.

As this string can't be generated, the question of ambiguity

Section D

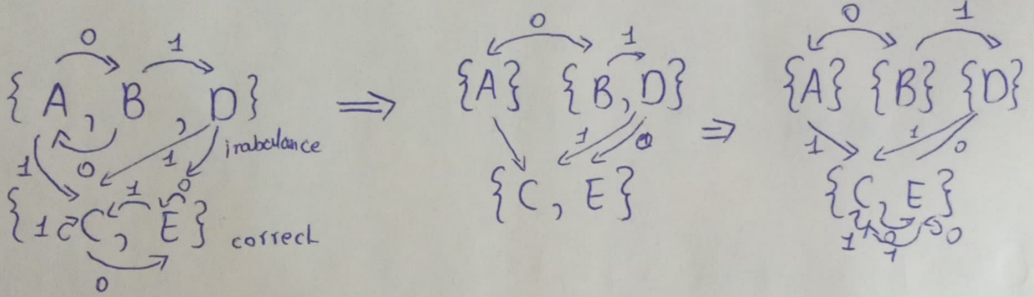
Q1)



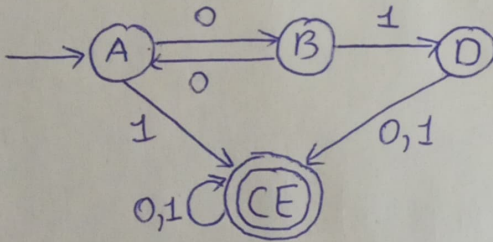
$\{A, B, D\}$ = Non-accepting

$\{C, E\}$ = accepting

Sol

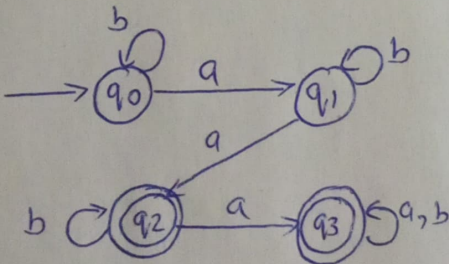


Minimised DFA



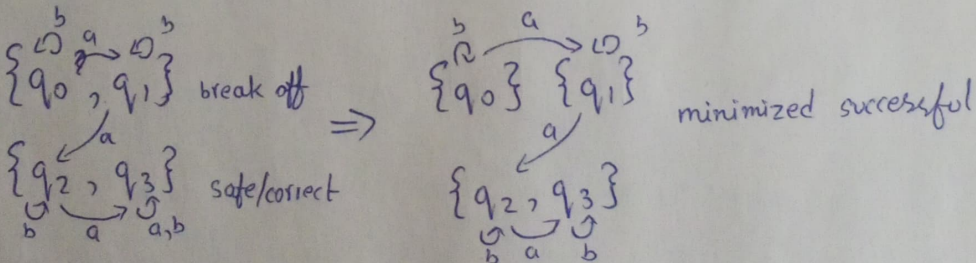
Q2)

DFA



Accepting = $\{q_2, q_3\}$

Non-accepting = $\{q_0, q_1\}$



Minimised DFA

